

Notes: Amiti and Davis (RES, 2012)

Trade, Firms, and Wages: Theory and Evidence

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1 Big picture

- **Question.** How does trade liberalization affect workers' wages when firms differ in whether they export final goods and import intermediate inputs?
- **Core idea.** Trade liberalization does not have one uniform wage effect. The wage effect depends on the worker's firm: domestic-only firms, exporters, importers, and firms that both export and import experience different changes in profits and therefore different wage changes.
- **Two distinct liberalizations.** The paper separates cuts in **output tariffs** from cuts in **input tariffs**. Output tariffs protect final-good producers from import competition. Input tariffs make imported intermediate inputs more expensive.
- **Main theoretical prediction.** A fall in output tariffs lowers wages at firms selling only domestically but can raise wages at exporters. A fall in input tariffs raises wages at firms that import inputs relative to firms that source inputs locally.
- **Empirical setting.** The paper studies Indonesian manufacturing from 1991 to 2000, a period with large output- and input-tariff cuts and detailed firm-level data on wages, exports, imported inputs, and individual intermediate input use.
- **Headline result.** A 10 percentage point fall in output tariffs reduces wages at domestic-only firms by roughly 3%, but can raise wages at sufficiently export-intensive firms. A 10 percentage point fall in input tariffs raises wages at high-import-share firms, with estimates as large as about 12% for firms importing all inputs.
- **Why this matters.** Globalization creates within-industry wage heterogeneity. The relevant unit is not just the industry or worker skill group, but the firm and its mode of global engagement.

2 Incremental contribution of the paper

- **Firm-level wage mechanism.** Prior trade-and-wage studies often focus on industry wages or skill groups. This paper puts firm heterogeneity at the center and explains why workers in the same industry may experience opposite wage effects.
- **Exports and imported inputs together.** The paper is among the first to model and estimate the joint wage consequences of output tariff cuts and input tariff cuts while allowing firms to export and import.
- **Intermediate-input tariffs.** A major empirical contribution is the construction of detailed input tariffs using firm-level information on individual intermediate inputs. This makes it possible to identify the wage effect of input liberalization separately from output liberalization.
- **Mode of globalization matters.** The paper shows that the same tariff cut has different effects depending on whether a firm is a non-globalizer, exporter, importer, or both exporter and importer.
- **Theory matched to data.** The model combines Melitz-style heterogeneous firms, imported intermediates, and firm-specific wages generated by a fair-wage constraint. The theory delivers directly testable interaction terms between tariffs and firm trade status.

- **Explanation for mixed prior findings.** Earlier studies found inconsistent wage effects of output tariff cuts. This paper shows that pooling exporters and non-exporters can obscure opposite firm-level effects.

3 Model setup

3.1 Consumers and final demand

Consumers have Dixit-Stiglitz preferences over differentiated final goods. The demand for variety ν is

$$q(\nu) = \left(\frac{p(\nu)}{P} \right)^{-\sigma} Q,$$

where $\sigma > 1$ is the elasticity of substitution, P is the aggregate price index, and Q is aggregate real consumption of the differentiated good.

Interpretation: The demand structure gives each firm a constant markup over marginal cost. Therefore, anything that changes marginal cost - such as importing cheaper or better intermediate inputs - changes prices, revenues, profits, and wages.

3.2 Fair-wage constraint

Workers are homogeneous, but firms may pay different wages. The paper assumes a fair-wage constraint:

$$W(0) = 1, \quad W_\nu = W(\pi_\nu), \quad 0 < W'(\pi_\nu) < \infty.$$

Zero-profit firms pay the outside wage normalized to one. More profitable firms pay higher wages.

Economic message: This is the bridge from trade shocks to firm wages. Trade liberalization first changes firm profits; the fair-wage relation then maps profit changes into wage changes.

3.3 Firms and modes of globalization

Potential final-good producers pay an entry cost, then draw three firm characteristics:

$$\lambda_\nu = (\phi_\nu, t_{M\nu}, t_{X\nu}),$$

where ϕ_ν is productivity, $t_{M\nu}$ is firm-specific import cost, and $t_{X\nu}$ is firm-specific export cost.

A firm can choose among four modes:

- produce for the domestic market only;
- import intermediate inputs only;
- export final goods only;
- both import intermediate inputs and export final goods.

The fixed costs associated with these modes are

$$F_\nu = \begin{cases} f, & \text{domestic only,} \\ f + nf_M, & \text{imports intermediate inputs,} \\ f + nf_X, & \text{exports final goods,} \\ f + n(f_X + f_M), & \text{imports and exports.} \end{cases}$$

Interpretation: Large, productive firms are more able to cover the fixed costs of exporting and importing. This generates the empirical prediction that exporters and importers are bigger, more productive, and higher wage.

3.4 Intermediate inputs and input tariffs

Final-good production uses labour and a composite of intermediate inputs. If a firm imports inputs, the effective price of imported inputs is

$$\tau_{M\nu} = \tau_M t_{M\nu},$$

where τ_M is the common input tariff or trade cost and $t_{M\nu}$ is firm-specific importing efficiency.

A cut in τ_M lowers marginal costs especially for firms that import intermediate inputs. This raises their profits and, through the fair-wage constraint, their wages.

3.5 Exports and output tariffs

Exporting gives firms access to foreign markets but requires fixed export costs and firm-specific iceberg export costs. A cut in output tariffs means tougher competition in the domestic final-goods market. Domestic-only firms lose protection, while exporters can benefit because their revenue base is larger and they are better positioned to expand.

4 Main theoretical predictions

- **Productivity cutoff rises with trade.** Moving from autarky to costly trade raises the survival cutoff. The least productive firms exit.
- **Domestic-only firms lose from final-good liberalization.** Output tariff reductions increase import competition and reduce profits and wages at domestic-only firms.
- **Large exporters gain from output tariff cuts.** Sufficiently export-intensive firms benefit from access to foreign demand and higher scale. Marginal exporters may still lose.
- **Importers gain from input tariff cuts.** Firms that import intermediate inputs experience lower marginal costs, higher profits, and higher wages.
- **Firm trade shares matter.** A firm with a larger export share or import share should have higher profits and wages, all else equal.
- **Synergy.** Exporting and importing can reinforce each other: exporters benefit more from input cost reductions, and importers may be better able to compete in export markets.

5 Data and institutional setting

5.1 Indonesian manufacturing census

The empirical analysis uses Indonesia’s manufacturing survey of large and medium-sized plants from 1991 to 2000. The survey covers firms with at least 20 employees, across 290 five-digit ISIC manufacturing industries.

Key firm-level variables include:

- total wage bill and employment, used to compute average firm wage;
- export value and export share;
- imported and domestically purchased intermediate inputs;
- ownership, employment, skill share, value added, and production data.

5.2 Tariff data

The paper constructs two separate tariff measures:

- **Output tariff:** the tariff on the firm’s final-good industry.
- **Input tariff:** a weighted average of tariffs on the intermediate inputs used by the industry.

A distinctive feature of the paper is the ability to build detailed input tariffs using firm-level information on individual inputs. From 1991 to 2000, average output tariffs fell from about 21% to 8%, while average input tariffs fell from about 14% to 6%.

Interpretation: The separation between output and input tariffs is essential. Output tariff cuts are pro-competitive shocks to final goods. Input tariff cuts are cost-reducing shocks for input importers.

5.3 Descriptive facts

The descriptive patterns are consistent with the model:

- only a minority of firms export, import, or both;
- globally engaged firms account for a disproportionate share of employment and value added;
- exporters, importers, and especially firms that both export and import pay higher wages;
- firms that both import and export are the largest and most productive.

6 Empirical strategy

6.1 Baseline firm-level wage equation

The main specification relates firm wages to tariffs and tariff interactions with firm trade status:

$$\ln(wage)_{f,i,t} = \alpha_f + \alpha_{l,t} + \beta_1 OutputTariff_{f,i,t} + \beta_2 OutputTariff_{f,i,t} \times FX_{f,i,t} \\ + \beta_3 InputTariff_{f,i,t} + \beta_4 InputTariff_{f,i,t} \times FM_{f,i,t} + Z'_{f,i,t} \Gamma + \varepsilon_{f,i,t}.$$

Here FX indicates that the firm exports, and FM indicates that the firm imports intermediate inputs. Firm fixed effects control for time-invariant firm heterogeneity; location-year effects control for region-specific shocks.

6.2 Predicted signs

- $\beta_1 > 0$: if output tariffs fall, wages at non-exporters fall.
- $\beta_2 < 0$: output tariff cuts affect exporters more positively than non-exporters.
- β_3 is expected to be positive in the simple model, though empirically it is often small or insignificant because non-importers may also benefit indirectly from cheaper domestic inputs.
- $\beta_4 < 0$: input tariff cuts raise wages at importers relative to non-importers.

6.3 Five-period differences and instruments

The authors also estimate five-period differences:

$$\Delta_5 \ln(wage)_{f,i,t} = \delta_1 \Delta_5 OutputTariff_{f,i,t} + \delta_2 \Delta_5 (OutputTariff_{f,i,t} \times FX_{f,i,t}) + \delta_3 \Delta_5 InputTariff_{f,i,t} + \delta_4 \Delta_5 (InputTariff_{f,i,t} \times FX_{f,i,t})$$

This helps reduce measurement error and absorb fixed firm heterogeneity. To address possible tariff endogeneity, the paper instruments tariff changes using initial industry characteristics and policy-related variables such as initial skill share, initial input tariffs, WTO exclusion indicators, and non-tariff barrier indicators.

7 Main empirical results

- **Output tariff cuts hurt domestic-only firms.** In the share specification, a 10 percentage point cut in output tariffs reduces wages in non-exporting firms by about 3%.
- **Output tariff cuts help sufficiently large exporters.** Firms exporting at least about half of output are estimated to experience wage gains from output tariff cuts. Firms exporting all output can see wage gains of about 3% from a 10 percentage point cut.
- **Input tariff cuts help importers.** Input tariff cuts raise wages at firms that import intermediate inputs. In the share specification, a 10 percentage point cut in input tariffs raises wages by up to about 12% for firms importing all inputs.
- **Effects differ within industries.** The same industry can contain domestic-oriented firms with predicted wage losses and globalized firms with predicted wage gains.
- **Results survive many checks.** The findings are robust to controls for ownership, skill share, firm size, productivity, alternative definitions of export/import status, selection corrections, exchange-rate controls, pre-crisis samples, alternative tariff measures, and dropping duty-free zones.

8 Mechanisms and interpretation

8.1 Profit channel

The model's mechanism is: tariff change $\rightarrow$ profit change $\rightarrow$ wage change. Because profits are not directly observed, the paper uses revenue as a proxy for firm performance. The results show that globalized firms' revenues rise relative to domestic-oriented firms after liberalization, consistent with the profit channel.

8.2 Skill composition

The paper checks whether wage effects are simply due to changes in worker composition. It uses production and non-production wages separately and, for 1995-1997, controls for detailed education shares. The tariff coefficients remain stable.

Interpretation: This does not prove that worker composition plays no role, but it shows that the main tariff-wage patterns are not mechanically driven by observed shifts in broad worker types or education categories.

8.3 Asian crisis

Because the sample covers the 1997-1998 Asian crisis, the authors test whether the results are driven by exchange rate changes or crisis-period distortions. They add trade-weighted real exchange rate interactions and also estimate the model on the pre-crisis period. The core tariff results remain.

9 Original figures and tables with reading notes

9.1 Figure 1: Determination of firm wage and profit for given mode of globalization and Table 1A: Importers, exporters and wages

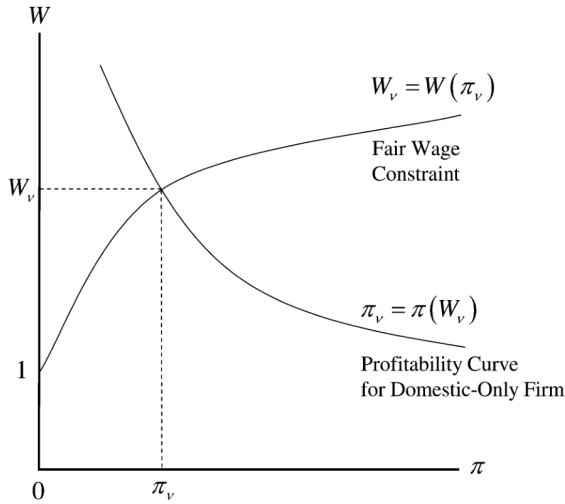


FIGURE 1

Determination of firm wage and profit for given mode of globalization

Figure 1: Determination of firm wage and profit for given mode of globalization

Read: Figure 1 shows how the fair-wage constraint and firm profitability curve jointly determine the firm wage. Table 1A documents that exporters, importers, and especially firms that both import and export pay higher wages than domestic-only firms.

Economic message: Figure 1 is the theoretical bridge from firm profits to firm wages. Table 1A shows why a firm-level theory is needed: wage premia are systematically related to firm globalization status.

Dependent variable	$\ln(\text{wage})_{f,i,t}$	$\ln(\text{wage})_{f,j,t}$	$\ln(\text{wage})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
Exporters	0.275*** (0.005)	0.176*** (0.005)	0.251*** (0.005)	0.161*** (0.005)	0.133*** (0.005)	0.076*** (0.005)
Importers	0.468*** (0.005)	0.245*** (0.005)	0.381*** (0.004)	0.214*** (0.004)	0.287*** (0.004)	0.146*** (0.004)
Importers and exporters	0.664*** (0.007)	0.445*** (0.006)	0.618*** (0.006)	0.422*** (0.006)	0.389*** (0.007)	0.254*** (0.007)
skillshare			1.367*** (0.013)	0.897*** (0.012)	1.279*** (0.013)	0.833*** (0.012)
$\ln(\text{labour})$					0.111*** (0.001)	0.097*** (0.001)
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	No	Yes	No	Yes	No	Yes
Firm effects	No	No	No	No	No	No
Observations	185,866	185,866	185,866	185,866	185,866	185,866
Adjusted R ²	0.30	0.52	0.37	0.54	0.39	0.55

Table 1A: Importers, exporters and wages

9.2 Table 1B: Importers, exporters and size and Table 2A: Tariffs and wages - baseline regressions

TABLE 1B
Importers, exporters and size

Dependent variable	$\ln(\text{labour})_{f,i,t}$	$\ln(\text{labour})_{f,i,t}$	$\ln(\text{VA})_{f,i,t}$	$\ln(\text{VA})_{f,i,t}$	$\ln(\text{TFP})_{f,i,t}$	$\ln(\text{TFP})_{f,i,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
Exporters	1.074*** (0.009)	0.889*** (0.009)	1.604*** (0.014)	1.297*** (0.014)	0.120*** (0.005)	0.107*** (0.005)
Importers	0.893*** (0.009)	0.731*** (0.009)	1.746*** (0.015)	1.261*** (0.014)	0.107*** (0.005)	0.116*** (0.004)
Importers and exporters	2.085*** (0.013)	1.749*** (0.012)	3.318*** (0.018)	2.692*** (0.018)	0.203*** (0.008)	0.202*** (0.006)
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry effects	No	Yes	No	Yes	No	Yes
Firm effects	No	No	No	No	No	No
Observations	185,866	185,866	172,235	172,235	153,018	153,018
Adjusted R ²	0.24	0.39	0.27	0.45	0.02	0.47

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 1B: Importers, exporters and size

TABLE 2A
Tariffs and wages—baseline regressions

Dependent variable: $\ln(\text{wage})_{f,i,t} - \ln(\text{wage})_{f,i,t-5}$					
Instrumental variables estimation					
	Output tariff (1)	With exporters (2)	Input tariffs (3)	With importers (4)	Both tariffs (5)
$\Delta \text{Output tariff}_{i,t}$	0.158 (0.184)	0.271 (0.186)			0.244 (0.187)
$\Delta (\text{Output tariff}_{i,t} \times \text{FX}_{f,i,t})$		-0.583*** (0.098)			-0.482*** (0.096)
$\Delta \text{Input tariff}_{i,t}$			-0.333* (0.190)	-0.209 (0.188)	-0.227 (0.196)
$\Delta (\text{Input tariff}_{i,t} \times \text{FM}_{f,i,t})$				-0.694*** (0.131)	-0.520*** (0.124)
$\Delta \text{FX}_{f,i,t}$	0.019*** (0.007)	0.129*** (0.019)	0.019*** (0.007)	0.022*** (0.007)	0.112*** (0.018)
$\Delta \text{FM}_{f,i,t}$	0.033*** (0.008)	0.031*** (0.008)	0.033*** (0.008)	0.112*** (0.016)	0.090*** (0.015)
Joint Significance tests Ho: sum of coefficients on tariff variables equals zero					
Output tariffs		-0.312** (0.154)			-0.238 (0.168)
Input tariffs				-0.903*** (0.217)	-0.748*** (0.222)
Weak instruments (F-stat)	2,501	1,818	22,000	8,515	1,273
Overidentification					
Hansen J statistic	5.97	5.51	0.28	5.82	4.90
p-value	0.05	0.06	0.60	0.12	0.09
Observations	55,393	55,393	55,393	55,393	55,393

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. Instruments include 1991 industry skill share, 1991 industry skill share interacted with five-period lagged export dummy, 1991 input tariff level, 1991 input tariff level interacted with five-period lagged import dummy, exclusion dummy = 1 if ten or more HS nine-digit products excluded within a five-digit industry code from commitment to reduce bound tariffs to 40%, and non-tariff dummy = 1 if ten or more HS nine-digit product codes were subject to non-tariff barriers. All of the estimations include location x year fixed effects.

Table 2A: Tariffs and wages - baseline regressions

Read: Table 1B shows that internationally engaged firms are larger and more productive. Table 2A provides the baseline tariff-wage regressions.

Interpretation: Table 2A establishes the paper's central empirical pattern: output tariff cuts affect exporters differently from non-exporters, and input tariff cuts affect importers differently from non-importers.

9.3 Table 2B: Tariffs and wages - additional controls, Part 1 and Table 2B: Tariffs and wages - additional controls, Part 2

	Dependent variable: $\ln(\text{wage})_{f,j,t} - \ln(\text{wage})_{f,j,t-5}$						
	Instrumental variables						OLS With trade shares (7)
	Without foreign firms (1)	With ownership (2)	With skill share (3)	With size (4)	With trade shares (5)	With trade bins (6)	
$\Delta \text{Output tariff}_{f,j}$	0.411** (0.185)	0.278 (0.187)	0.329* (0.185)	0.316* (0.186)	0.334* (0.187)	0.341* (0.186)	0.144* (0.082)
$\Delta \text{Output tariff}_{f,j} \times \text{FX}_{f,j,t}$	-0.440*** (0.102)	-0.485*** (0.096)	-0.402*** (0.096)	-0.553*** (0.097)	-0.646*** (0.138)	-0.427*** (0.115)	-0.338*** (0.080)
$\Delta \text{Output tariff}_{f,j} \times \text{highFX}_{f,j,t}$					-0.250* (0.128)		
$\Delta \text{Input tariff}_{f,j}$	-0.261 (0.197)	-0.242 (0.197)	-0.244 (0.197)	-0.232 (0.194)	-0.254 (0.196)	-0.230 (0.197)	-0.074 (0.196)
$\Delta \text{Input tariff}_{f,j} \times \text{FM}_{f,j,t}$	-0.528*** (0.134)	-0.540*** (0.125)	-0.551*** (0.125)	-0.598*** (0.129)	-1.148*** (0.307)	-0.794 (0.242)	-0.790*** (0.184)
$\Delta \text{Input tariff}_{f,j} \times \text{highFM}_{f,j,t}$					-0.452* (0.244)		
$\Delta \text{FX}_{f,j,t}$	0.104*** (0.020)	0.110*** (0.019)	0.110*** (0.018)	0.127*** (0.018)	0.147*** (0.027)	0.107*** (0.021)	0.085*** (0.017)
$\Delta \text{FM}_{f,j,t}$	0.082*** (0.016)	0.092*** (0.015)	0.091*** (0.016)	0.104*** (0.016)	0.179*** (0.041)	0.053* (0.029)	0.137*** (0.025)
$\Delta \text{Skill share}_{f,j,t}$					0.283*** (0.029)	0.286*** (0.028)	0.286*** (0.028)
$\Delta \ln(\text{labour})_{f,j,t}$					-0.065*** (0.007)	-0.064*** (0.007)	-0.065*** (0.007)
$\Delta \text{foreign share}_{f,j,t}$		0.111*** (0.027)	0.116*** (0.027)	0.126*** (0.027)	0.127*** (0.028)	0.126*** (0.028)	0.130*** (0.028)
$\Delta \text{Govt share}_{f,j,t}$		0.067*** (0.014)	0.066*** (0.014)	0.065*** (0.014)	0.067*** (0.014)	0.066*** (0.014)	0.064*** (0.014)

(continued)

	Dependent variable: $\ln(\text{wage})_{f,j,t} - \ln(\text{wage})_{f,j,t-5}$						
	Instrumental variables						OLS With trade shares (7)
	Without foreign firms (1)	With ownership (2)	With skill share (3)	With size (4)	With trade shares (5)	With trade bins (6)	
Joint significance tests (the sum of coefficients on tariff variables equals zero)							
Output tariffs	-0.029 (0.168)	-0.207 (0.168)	-0.163 (0.168)	-0.217 (0.169)	-0.312* (0.182)	-0.336* (0.179)	-0.194* (0.101)
Input tariffs	-0.789*** (0.224)	-0.783*** (0.223)	-0.775*** (0.221)	-0.830*** (0.224)	-1.402*** (0.350)	-0.877*** (0.229)	-0.864*** (0.218)
Hansen J statistic	1.94	4.69	4.16	4.21	3.43	4.50	
p-value	0.38	0.10	0.12	0.12	0.18	0.11	
Observations	52,383	55,393	55,393	55,393	55,393	55,393	55,393
Adjusted R ²							0.04

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. Instruments include 1991 industry skill share, 1991 industry skill share interacted with five-period lagged export dummy, 1991 input tariff level, 1991 input tariff level interacted with five-period lagged import dummy, exclusion dummy = 1 if ten or more HS nine-digit products excluded within a five-digit industry code from commitment to reduce bound tariffs to 40%, and non-tariff dummy = 1 if ten or more HS nine-digit product codes were subject to non-tariff barriers. All specifications pass the weak instrument test with F-stat over 1,000 and Column 6 also includes $\Delta \text{highFX}_{f,j,t}$ and $\Delta \text{highFM}_{f,j,t}$, but are not reported to save space. All the estimations include location-year fixed effects.

Table 2B: Tariffs and wages - additional controls, Part 2

Table 2B: Tariffs and wages - additional controls, Part 1

Read: Table 2B adds controls and alternative interaction structures. It includes ownership, skill share, firm size, trade shares, high-trade-share bins, and an OLS comparison.

Takeaway: The main coefficients are stable. The share specification reveals the key threshold logic: small or marginal exporters may not gain, but sufficiently export-intensive firms do.

9.4 Table 3A: Tariffs and wages - heterogeneity and selection, Part 1 and Table 3A: Tariffs and wages - heterogeneity and selection, Part 2

	Dependent variable: $\ln(\text{wage})_{f,j,t} - \ln(\text{wage})_{f,j,t-5}$					
	Instrumental variables estimation					
	Alternative definitions of global status					
With productivity (1)	Any time in sample (2)	As in 1991 (3)	At entry (4)	At entry; drop switchers (5)	At entry; with shares (6)	
$\Delta \text{Output tariff}_{f,j}$	0.374** (0.157)	0.405*** (0.156)	0.316* (0.172)	0.401*** (0.157)	0.513*** (0.170)	0.400*** (0.159)
$\Delta \text{Output tariff}_{f,j} \times \text{FX}_{f,j,t}$	-0.435*** (0.087)	-0.424*** (0.086)	-0.407*** (0.130)	-0.574*** (0.115)	-0.509*** (0.127)	-0.601*** (0.163)
$\Delta \text{Input tariff}_{f,j}$	-0.141 (0.171)	-0.055 (0.170)	-0.025 (0.189)	-0.111 (0.171)	-0.185 (0.181)	-0.135 (0.172)
$\Delta \text{Input tariff}_{f,j} \times \text{FM}_{f,j,t}$	-0.514*** (0.129)	-0.448*** (0.136)	-0.613*** (0.173)	-0.563*** (0.151)	-0.354** (0.159)	-1.039*** (0.327)
$\Delta \text{FX}_{f,j,t}$	0.039*** (0.009)	0.038** (0.007)	0.039** (0.009)	0.030*** (0.008)	0.037*** (0.012)	0.037*** (0.012)
$\Delta \text{FM}_{f,j,t}$	0.032*** (0.008)	0.038** (0.008)	0.039** (0.008)	0.028*** (0.008)	0.032** (0.018)	0.032** (0.018)
$\Delta \text{Skill share}_{f,j,t}$	0.251*** (0.025)	0.232*** (0.025)	0.253*** (0.026)	0.282*** (0.025)	0.257*** (0.025)	0.282*** (0.025)
$\Delta \ln(\text{labour})_{f,j,t}$	-0.042*** (0.006)	-0.044*** (0.006)	-0.059*** (0.007)	-0.042*** (0.006)	-0.046*** (0.007)	-0.042*** (0.006)
$\Delta \text{foreign share}_{f,j,t}$	0.083*** (0.026)	0.082*** (0.026)	0.084 (0.031)	0.085*** (0.026)	0.080*** (0.032)	0.080*** (0.027)
$\Delta \text{Govt share}_{f,j,t}$	0.080*** (0.014)	0.082*** (0.014)	0.076*** (0.015)	0.081*** (0.014)	0.084*** (0.017)	0.082*** (0.014)
$\Delta \ln(\text{VA per worker})_{f,j,t}$	0.179*** (0.005)	0.179*** (0.005)	0.172*** (0.006)	0.180*** (0.005)	0.180*** (0.006)	0.180*** (0.005)

(continued)

	Dependent variable: $\ln(\text{wage})_{f,j,t} - \ln(\text{wage})_{f,j,t-5}$					
	Instrumental variables estimation					
	With productivity (1)	Any time in sample (2)	As in 1991 (3)	At entry (4)	At entry; drop switchers (5)	At entry; with shares (6)
Joint Significance tests (the sum of coefficients on tariff variables equals zero)						
Output tariffs	-0.060 (0.140)	-0.019 (0.143)	-0.092 (0.158)	-0.173 (0.153)	0.004 (0.176)	-0.292* (0.176)
Input tariffs	-0.655*** (0.205)	-0.504*** (0.120)	-0.637*** (0.237)	-0.674*** (0.215)	-0.539*** (0.221)	-1.175*** (0.355)
Hansen J statistic	3.19	3.08	1.89	2.82	0.57	2.36
p-value	0.20	0.21	0.39	0.24	0.75	0.31
Observations	50,484	50,484	38,335	50,484	38,275	50,484

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. Instruments include 1991 industry skill share, 1991 industry skill share interacted with five-period lagged export dummy, 1991 input tariff level, 1991 input tariff level interacted with five-period lagged import dummy, exclusion dummy = 1 if ten or more HS nine-digit products excluded within a five-digit industry code from commitment to reduce bound tariffs to 40%, and non-tariff dummy = 1 if ten or more HS nine-digit product codes were subject to non-tariff barriers. All specifications pass the weak instrument test with F-stat over 800. All the estimations include location-year fixed effects.

Table 3A: Tariffs and wages - heterogeneity and selection, Part 2

Table 3A: Tariffs and wages - heterogeneity and selection, Part 1

Read: Table 3A controls for productivity and tests alternative fixed definitions of firms' global status: ever exporter/importer, 1991 status, status at entry, dropping switchers, and trade shares at entry.

Interpretation: The results are not driven only by firms switching into importing or exporting after tariff cuts. The tariff interactions remain important even when global status is fixed in several

ways.

9.5 Table 3B: Tariffs and wages - heterogeneity and selection, Part 1 and Table 3B: Tariffs and wages - heterogeneity and selection, Part 2

TABLE 3B
Tariffs and wages—heterogeneity and selection

	Dependent variable: $\ln(wage)_{f,j,t}$					
	baseline	1 st stage probit	Heckman	heterogeneity	non-parametric 1 st stage	With exit
	(1)	(2)	(3)	(4)	(5)	(6)
Output tariff _{<i>f,j</i>}	0.021 (0.045)	0.355** (0.169)	0.011 (0.049)	0.006 (0.048)	0.462*** (0.170)	0.015 (0.045)
Output tariff _{<i>f</i>} x FX _{<i>f,j,t</i>}	-0.096*** (0.043)		-0.095** (0.044)	-0.089** (0.043)	-0.100** (0.049)	-0.091** (0.043)
Input tariff _{<i>f</i>}	0.120 (0.099)	-1.332*** (0.463)	0.159 (0.123)	0.174 (0.123)	-1.372*** (0.454)	0.177 (0.132)
Input tariff _{<i>f</i>} x FM _{<i>f,j,t</i>}	-0.385*** (0.083)		-0.382*** (0.081)	-0.368*** (0.082)	-0.412*** (0.019)	-0.370*** (0.083)
Skill share _{<i>f,j,t</i>}	0.297*** (0.021)	-0.202*** (0.044)	0.300*** (0.022)	0.301*** (0.022)	-0.139 (0.102)	0.296*** (0.021)
$\ln(\text{labour})_{f,j,t}$	-0.068*** (0.006)	0.125*** (0.007)	-0.072*** (0.007)	-0.074*** (0.007)	-0.076*** (0.007)	-0.070*** (0.006)
$\ln(\text{VA per worker})_{f,j,t}$	0.143*** (0.004)	0.028*** (0.006)	0.142*** (0.004)	0.142*** (0.004)	0.142*** (0.004)	0.142*** (0.004)
Age _{<i>f,j,t</i>}	0.004*** (0.001)	-0.002 (0.001)	-0.002 (0.0004)	-0.004 (0.0004)	-0.002 (0.0004)	-0.002 (0.0004)
Age _{<i>f,j,t</i>} x Finance dependence _{<i>f</i>}	-0.004*** (0.001)	-0.001 (0.0007)	-0.001 (0.0007)	-0.003 (0.0007)	-0.003** (0.001)	-0.00094 (0.0008)

(continued)

Table 3B: Tariffs and wages - heterogeneity and selection, Part 1

TABLE 3B
Continued

	Dependent variable: $\ln(wage)_{f,j,t}$					
	baseline	1 st stage probit	Heckman	heterogeneity	nonparametric 1 st stage	With exit
	(1)	(2)	(3)	(4)	(5)	(6)
Exit _{<i>f,j,t</i>} =1 if exit in t+1						-0.043*** (0.006)
IMR _{<i>f,j,t</i>}			-0.924*** (0.256)			
Series terms: <i>p</i> -values from <i>F</i> -tests						
$\Delta \ln(wage)_{f,j,t}$ (3 rd order)				0.000***		
Age _{<i>f,j,t</i>} (4th order)					0.004***	
$\ln(\text{VA per worker})_{f,j,t}$ (3 rd order)					0.000***	
$\ln(\text{labour})_{f,j,t}$					0.000***	
Interaction terms						0.000***
Propensity _{<i>f,j,t</i>} (3 rd order)						0.000***
Joint significance tests: The sum of coefficients on tariff variables equals zero						
Output tariff	-0.074 (0.056)		-0.084 (0.063)	-0.083 (0.062)		-0.084 (0.061)
Input tariff	-0.264** (0.116)		-0.223* (0.132)	-0.194 (0.134)		-0.235* (0.142)
Observations	109,302	125,962	109,302	109,302	125,962	107,270
Adjusted R ²	0.86	0.02	0.86	0.86	0.02	0.85

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. Standard errors in Columns 3, 4, and 6 have been bootstrapped with 200 replications. All the estimations also include FX, FM, foreign share, and government share but the coefficients are suppressed to save space. The top and bottom one percentiles of the propensity measure have been trimmed. All estimations have location-year effects and firm-fixed effects except the first-stage probits in Columns 2 and 5, which include year and industry effects.

Table 3B: Tariffs and wages - heterogeneity and selection, Part 2

Read: Table 3B studies selection and heterogeneity bias in levels specifications. It uses Heckman corrections, non-parametric selection corrections, and an exit control.

Economic message: The exit result is intuitive: firms about to exit pay lower wages. But selection corrections do not eliminate the main tariff-wage patterns.

9.6 Table 4: Asian crisis and alternative tariff measures, Part 1 and Table 4: Asian crisis and alternative tariff measures, Part 2

TABLE 4
Asian crisis and alternative tariff measures

	Dependent variable: $\ln(wage)_{f,j,t}$						
	Asian crisis						
	With exchange rates	1991–1996	Input tariffs (EO 1995)	Input tariffs (EO 1998)	Firm level tariffs	Weighted tariffs	Drop firms in Duty-free zones
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Output tariff _{<i>f</i>}	0.000* (0.051)	0.083 (0.061)	0.090** (0.039)	0.099** (0.040)	0.078** (0.035)	0.055* (0.051)	0.098** (0.051)
Output tariff _{<i>f</i>} x FX _{<i>f,j,t</i>}	-0.116*** (0.046)	-0.122*** (0.045)	-0.210*** (0.043)	-0.204*** (0.042)	-0.165*** (0.042)	-0.111*** (0.041)	-0.203*** (0.043)
Input tariff _{<i>f</i>}	-0.029 (0.095)	0.086 (0.154)	-0.069 (0.054)	-0.127* (0.072)	0.180*** (0.062)	-0.071* (0.041)	-0.034 (0.096)
Input tariff _{<i>f</i>} x FM _{<i>f,j,t</i>}	-0.402*** (0.094)	-0.434*** (0.123)	-0.159*** (0.047)	-0.238*** (0.071)	-0.302*** (0.065)	-0.427*** (0.082)	-0.454*** (0.085)
FX _{<i>f,j,t</i>}	0.031*** (0.011)	0.066*** (0.014)	0.060*** (0.009)	0.057*** (0.009)	0.048*** (0.009)	0.030*** (0.008)	0.057*** (0.009)
FM _{<i>f,j,t</i>}	0.097*** (0.013)	0.077*** (0.017)	0.055*** (0.007)	0.059*** (0.007)	0.074*** (0.009)	0.081*** (0.009)	0.090*** (0.011)
Skill share _{<i>f,j,t</i>}	0.278*** (0.020)	0.260*** (0.024)	0.278*** (0.020)	0.278*** (0.020)	0.303*** (0.022)	0.279*** (0.020)	0.276*** (0.020)
$\Delta \ln(\text{labour})_{f,j,t}$	0.072*** (0.006)	0.073*** (0.009)	0.071*** (0.006)	0.071*** (0.006)	0.068*** (0.006)	0.071*** (0.006)	0.073*** (0.006)
Foreign share _{<i>f,j,t</i>}	0.130*** (0.018)	0.111*** (0.026)	0.132*** (0.018)	0.132*** (0.018)	0.149*** (0.019)	0.129*** (0.018)	0.126*** (0.018)

(continued)

Table 4: Asian crisis and alternative tariff measures, Part 1

TABLE 4
Continued

	Dependent variable: $\ln(wage)_{f,j,t}$						
	Asian crisis						
	With exchange rates	1991–1996	Input tariffs (EO 1995)	Input tariffs (EO 1998)	Firm level tariffs	Weighted tariffs	Drop firms in Duty-free zones
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\Delta \ln(wage)_{f,j,t}$	0.059*** (0.010)	0.034** (0.018)	0.077*** (0.010)	0.057*** (0.010)	0.069*** (0.010)	0.057*** (0.010)	0.058*** (0.010)
Exit _{<i>f,j,t</i>} if exit in t+1	-0.047*** (0.006)	-0.039*** (0.006)	-0.048*** (0.006)	-0.048*** (0.006)	-0.049*** (0.011)	-0.048*** (0.006)	-0.048*** (0.006)
TWRR _{<i>f</i>} x FX _{<i>f,j,t</i>}	-0.096*** (0.021)						
TWRR _{<i>f</i>} x FM _{<i>f,j,t</i>}	0.012 (0.019)						
Joint significance tests: The sum of coefficients on tariff variables equals zero							
Output tariffs	-0.025 (0.059)	-0.046 (0.073)	-0.129*** (0.049)	-0.105** (0.049)	-0.087* (0.052)	-0.056 (0.045)	-0.105* (0.058)
Input tariffs	-0.521*** (0.175)	-0.349** (0.166)	-0.228*** (0.055)	-0.365*** (0.088)	-0.123* (0.070)	-0.498*** (0.090)	-0.488*** (0.113)
Observations	185,866	105,262	185,866	185,866	126,576	184,928	184,809
Adjusted R ²	0.82	0.82	0.82	0.82	0.80	0.82	0.82

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. All the estimations include firm-fixed effects and location-year fixed effects.

Table 4: Asian crisis and alternative tariff measures, Part 2

Read: Table 4 checks whether the Asian crisis, exchange rates, alternative input-tariff weights, firm-level input tariffs, import-weighted tariffs, or duty-free zones drive the results.

Interpretation: The coefficients remain stable. This is important because the sample includes the 1997-1998 crisis and because input tariffs require reconstruction from input-use data.

9.7 Table 5: Mechanisms, Part 1 and Table 5: Mechanisms, Part 2

Dependent variable	$\ln(\text{revenue})_{f,i,t}$	$\ln(\text{wu})_{f,i,t}$	$\ln(\text{ws})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$
				1995–1997	1995–1997	1995–1997
				With skill share	With education share	
	(1)	(2)	(3)	(4)	(5)	(6)
Output tariff _{<i>f,i,t</i>}	−0.028 (0.076)	0.119** (0.051)	0.134*** (0.053)	0.452*** (0.132)	0.460*** (0.131)	0.463*** (0.130)
Output tariff _{<i>f,i,t</i>} x FX _{<i>f,i,t</i>}	−0.399*** (0.072)	−0.202*** (0.045)	−0.147*** (0.057)	−0.277*** (0.099)	−0.272*** (0.098)	−0.264*** (0.094)
Input tariff _{<i>f,i,t</i>}	0.130 (0.168)	−0.085 (0.097)	−0.014 (0.099)	−0.529* (0.310)	−0.516* (0.306)	−0.524* (0.302)
Input tariff _{<i>f,i,t</i>} x FM _{<i>f,i,t</i>}	−0.649*** (0.130)	−0.573*** (0.095)	−0.225** (0.100)	−0.600*** (0.215)	−0.613*** (0.215)	−0.600*** (0.203)
FX _{<i>f,i,t</i>}	0.148*** (0.015)	0.047*** (0.010)	0.072*** (0.011)	0.069*** (0.018)	0.068*** (0.018)	0.065*** (0.017)
FM _{<i>f,i,t</i>}	0.251*** (0.019)	0.092*** (0.012)	0.077*** (0.014)	0.100*** (0.027)	0.100*** (0.027)	0.098*** (0.026)
skillshare _{<i>f,i,t</i>}	0.050 (0.034)	0.570*** (0.020)	−1.595*** (0.033)		0.270*** (0.042)	
$\Delta \ln(\text{labour})_{f,i,t}$	0.794*** (0.016)	−0.062*** (0.006)	0.002*** (0.006)	−0.128*** (0.014)	−0.126*** (0.014)	−0.121*** (0.014)
Exit _{<i>f,i,t</i>} if exit in t+1	−0.082*** (0.009)	−0.052*** (0.006)	−0.026*** (0.008)	−0.054*** (0.009)	−0.054*** (0.009)	−0.053*** (0.008)
Education shares _{<i>f,i,t</i>}						−0.938***
Production_1						−0.929***
Production_2						−0.890***
Production_3						−0.835***
Production_4						−0.599***
Production_5						−0.831***
Non-production_1						−0.926***
Non-production_2						−0.669***
Non-production_3						−0.446***
Non-production_4						

(continued)

Table 5: Mechanisms, Part 1

Read: Table 5 investigates mechanisms. It examines revenue, skilled and unskilled wages, and specifications with skill-share and education-share controls.

Economic message: The table supports the idea that globalized firms improve performance after liberalization, and it reduces the concern that the wage findings are merely compositional shifts across observed skill categories.

Dependent Variable	$\ln(\text{revenue})_{f,i,t}$	$\ln(\text{wu})_{f,i,t}$	$\ln(\text{ws})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$	$\ln(\text{wage})_{f,i,t}$
					1995–1997	
					With skill share	With education share
	(1)	(2)	(3)	(4)	(5)	(6)
Joint significance tests Ho: sum of coefficients on tariff variables equals zero						
Output tariffs	−0.427*** (0.095)	−0.083 (0.060)	−0.013 (0.070)	0.175 (0.142)	0.188 (0.141)	0.199 (0.140)
Input tariffs	−0.519*** (0.192)	−0.658*** (0.118)	−0.239* (0.132)	−1.129*** (0.311)	−1.129*** (0.308)	−1.123*** (0.304)
Observations	173,732	185,795	149,575	61,901	61,901	61,901
Adjusted R ²	0.93	0.79	0.71	0.85	0.85	0.85

Notes: Robust standard errors in parentheses; * significant at 10%; ** significant at 5%; *** significant at 1%. Education shares are defined as the number of workers in each category divided by total employment. The categories are 1–5: 1—“not finished primary school”; 2—primary school; 3—junior high school; 4—senior high school; 5—diploma or higher. All the estimations also include foreign share and government share, but the coefficients are suppressed to save space. All the estimations include firm-fixed effects and location-year effects.

rising to 88% in the course of the shock. Export growth was thus very highly quality biased. The case of Indonesia is quite distinct. Indonesia is one of the least skill-abundant countries

Table 5: Mechanisms, Part 2

9.8 Figure 2: Modes of globalization and firm-wage changes

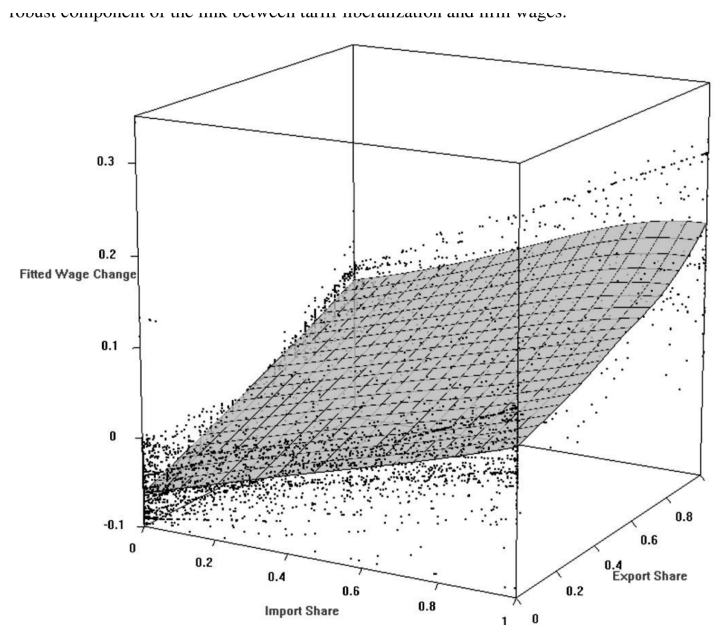


FIGURE 2
Modes of globalization and firm-wage changes. The predominance of negative values near the origin indicates that

Figure 2: Modes of globalization and firm-wage changes

Read: Figure 2 plots fitted wage changes as a function of import share and export share.

Interpretation: The negative fitted wage changes are concentrated near the origin, where firms are domestic-oriented. The positive fitted wage changes are concentrated away from the origin, where firms have higher import or export shares.

9.9 Table A1: Summary statistics, Part 1 and Table A1: Summary statistics, Part 2

TABLE A1
Summary statistics

Variable	Levels		Five-period differences	
	Mean	Standard deviation	Mean	Standard deviation
$\ln(\text{wage})_{f,i,t}$	7.35	0.80	0.70	0.54
$\ln(\text{wsk})_{f,i,t}$	7.87	0.92		
$\ln(\text{wob})_{f,i,t}$	7.25	0.79		
$\ln(\text{labour})_{f,i,t}$	4.18	1.19	0.01	0.53
Skill share $f_{i,t}$	0.14	0.15	0.00	0.14
Foreign share $f_{i,t}$	0.04	0.18	0.00	0.11
Govt share $f_{i,t}$	0.02	0.14	0.17	0.40
Export share $f_{i,t}$	0.11	0.29	-0.02	0.28
Export share $f_{i,t}$ if FX=1	0.71	0.33		
FX $f_{i,t}$ =1 if exshare>0	0.15	0.36	-0.03	0.38
Import share $f_{i,t}$	0.09	0.24	-0.01	0.20
Import share $f_{i,t}$ if FM=1	0.47	0.36		
FM $f_{i,t}$ =1 if import share >0	0.19	0.39	-0.02	0.34
Exit $f_{i,t}$ =1 if exit next period	0.08	0.28		
Age $f_{i,t}$	12.67	12.73		
Output tariff $f_{i,t}$	0.17	0.11	-0.12	0.08
Input tariff $f_{i,t}$	0.11	0.06	-0.03	0.08
$\ln(\text{TFP})_{f,i,t}$	1.63	0.66		
$\ln(\text{VA per worker})_{f,i,t}$	8.56	1.25	0.67	0.96
$\ln(\text{revenue})_{f,i,t}$	13.62	2.06		
Financial dependence	0.23	0.35		
Production_1 $f_{i,t}$	0.08	0.17		
Production_2 $f_{i,t}$	0.36	0.29		
Production_3 $f_{i,t}$	0.22	0.20		
Production_4 $f_{i,t}$	0.19	0.22		
Production_5 $f_{i,t}$	0.01	0.03		
Non-production_1 $f_{i,t}$	0.00	0.03		

(continued)

TABLE A1
Continued

Variable	Levels		Five-period differences	
	Mean	Standard deviation	Mean	Standard deviation
Non-production_2 $f_{i,t}$	0.02	0.06		
Non-production_3 $f_{i,t}$	0.02	0.05		
Non-production_4 $f_{i,t}$	0.07	0.09		
Non-production_5 $f_{i,t}$	0.02	0.04		
Observations	185,866		55,393	

Table A1: Summary statistics, Part 2

Table A1: Summary statistics, Part 1

Read: Table A1 reports summary statistics for firm wages, trade shares, ownership, employment, tariffs, productivity, education shares, and the five-period differenced sample.

Takeaway: The summary statistics clarify the empirical environment: export and import participation are limited but economically important, and tariff changes are large enough to identify meaningful wage effects.

10 How to read the paper's contribution in one pass

- Start with the model's fair-wage diagram. It says that firm wage differences can arise from firm profit differences.
- Then separate two shocks: output-tariff cuts and input-tariff cuts.
- Output-tariff cuts are mainly competition shocks for domestic producers, but exporters are better positioned to benefit.
- Input-tariff cuts are cost shocks that help firms that import intermediate inputs.
- The empirical tables show that these predictions appear in Indonesian plant data: tariff cuts raise wages at firms most connected to global markets and reduce or fail to raise wages at firms oriented domestically.

11 Limitations and open questions

- **Worker-level data.** The paper uses firm-level average wages, not matched employer-employee data. This limits the ability to fully separate firm wage premia from unobserved worker composition.

- **Profit measurement.** The theory runs through profits, but the data do not provide a clean profit measure. Revenue and robustness tests offer indirect evidence.
- **Foreign ownership.** The paper controls for foreign ownership and drops foreign firms in robustness checks, but it does not build a full theory of multinational ownership.
- **General equilibrium wage effects.** Year effects absorb economy-wide wage changes, so the estimates are best interpreted as relative firm-level effects.
- **Mechanism competition.** Fair wages are one plausible channel. Search-and-bargaining, efficiency wages, quality upgrading, and worker sorting may also matter.

12 Bottom line

Amiti and Davis show that the wage effects of trade liberalization depend sharply on firm global engagement. Output tariff cuts hurt workers at domestic-only firms but can benefit workers at exporters. Input tariff cuts benefit workers at firms that import intermediate inputs. The paper's core lesson is that trade liberalization redistributes not only across industries and skill groups, but also across firms within narrowly defined industries.